

The Interrelationship between AI, ML, and Data Science: Overlap vs. Independence

Research Review

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Abstract

Artificial Intelligence (AI), Machine Learning (ML), and Data Science are three interrelated fields that often intersect but also differ in purpose and scope. Machine learning focuses on enabling algorithms to infer patterns from datasets, data science covers the end-to-end workflow of collecting, preparing, and interpreting data, and artificial intelligence aims to design systems that can reason or act autonomously. While there is overlap in areas such as predictive modeling, automation, and data-driven decision support, each discipline preserves its distinctive objectives, methodologies, and applications. A thorough understanding of their connections and distinctions is essential to appreciate how these domains individually and collectively contribute to contemporary technological progress.

Keywords

Artificial Intelligence; Machine Learning; Data Science; Overlap; Differences; Data Analysis; Algorithms; Predictive Models; Automation.

Introduction

Artificial Intelligence (AI), Machine Learning (ML), and Data Science have emerged as three foundational disciplines shaping modern technological ecosystems. Although often used interchangeably, each field holds its own purpose, methodology, and scope while simultaneously sharing strong interconnections. Their relationship can be understood as a layered framework: **Data Science** focuses on extracting insights from data, **ML** develops algorithms that learn from data, and **AI** aims to build systems capable of intelligent, autonomous behavior. Understanding how these domains overlap yet remain independent is crucial for interpreting current advancements in automation, analytics, and decision-making technologies.

Artificial Intelligence (AI), the oldest of the three fields, dates back to the mid-20th century with the ambition of replicating human-like reasoning, problem-solving, and perception. Over time, AI evolved from symbolic logic systems to probabilistic models and, eventually, to learning-driven approaches that rely heavily on ML.

Machine Learning (ML), introduced formally through Mitchell's definition—where a system improves its performance on a task through experience [12]—shifted the focus from programmed rules to data-driven adaptation. This transition laid the foundation for modern techniques like neural networks, reinforcement learning, and deep learning, which now power applications in vision, speech processing, recommendation systems, and autonomous technology.

Data Science, meanwhile, integrates statistical analysis, computational tools, and domain expertise to handle large and complex datasets. Its role encompasses data cleaning, feature engineering, visualization, and interpretation [2]—activities essential for building reliable ML and AI systems. Donoho's "**Greater Data Science**" [2] framework highlights how Data Science acts as both the starting point and analytical backbone for AI-powered applications.

Despite significant overlap, these fields maintain distinct goals. Data Science emphasizes insight generation and decision support, ML focuses on predictive modeling and pattern discovery, while AI aims for adaptable, interactive, and often autonomous systems. Their synergy is most evident in modern industries where data pipelines feed ML models, and ML models enable AI-driven automation. This literature review examines how these three domains intersect, support one another, and maintain

independence in methodologies and applications, providing a clear understanding of their roles in today's technological landscape.

Literature Review

Artificial Intelligence (AI), Machine Learning (ML), and Data Science represent three closely related yet distinct fields that form the foundation of modern data-driven technological advancements. Each domain contributes uniquely to the processing, analysis, and utilization of data, and understanding their definitions and interrelationships is crucial for navigating contemporary developments. This literature review explores the core principles and distinctions of these fields, emphasizing both their overlaps and areas of independence, supported by seminal academic contributions.

AI is the broadest of the three, aspiring to endow machines with human-like cognitive abilities such as reasoning, learning, perception, and autonomous decision-making. It traces its roots back to the mid-20th century, initially focusing on symbolic logic and computational representations of intelligence. Over time, AI migrated towards probabilistic models and, more recently, to learning-based models primarily driven by advances in ML. Despite significant progress, Jordan and Mitchell (2015) emphasize that AI is still largely narrow, excelling in specialized tasks but lacking general intelligence akin to humans. [1] Modern AI applications are diversified, encompassing robotics, natural language processing, computer vision, and strategic game playing, all leveraging a mixture of

rule-based systems and deep learning methodologies to interpret and interact with the environment. These advances highlight AI's primary goal of creating systems capable of adaptive, context-aware, and autonomous operation beyond simple data processing. As AI evolved from symbolic reasoning toward computational models capable of learning from data, Machine Learning emerged as the key driver enabling these intelligent behaviors. This shift makes ML a natural next focus, as it provides the statistical and algorithmic foundations that power most modern AI systems.

ML, as a specialized subset of AI, focuses on algorithms that improve their performance on specific tasks through experience, formalized by Mitchell's learning framework, where performance **P** on task **T** improves with experience **E** [12]. This data-centric approach has revolutionized computational intelligence by enabling systems to learn patterns from data instead of relying on rigid programming. Key paradigms within ML include:

- Supervised learning — where models learn from labeled data for tasks like classification and regression
- Unsupervised learning — that discovers hidden data structures such as clusters or reduced-dimensional representations, and
- Reinforcement learning — which emphasizes sequential

decision-making through trial-and-error interactions with the environment.

- Semi-supervised and Self-supervised learning — more modern approaches that address scenarios with limited labeled data, supporting scalable and robust learning paradigms critical in domains like natural language processing and computer vision.

Theoretical foundations by Shalev-Shwartz and Ben-David (2014) ensure these models generalize beyond the training data [6], while Kelleher et al. (2015) stress their applications across industries ranging from healthcare to finance [5]. Since ML systems rely heavily on well-prepared, meaningful, and high-quality data, the role of Data Science becomes essential. Data Science supports ML by managing data collection, cleaning, exploration, and interpretation—creating the analytical environment in which ML models are developed, validated, and applied. This makes Data Science the next logical component in understanding the broader technological landscape.

Data Science, distinct yet symbiotic with AI and ML, combines statistics, computing, and domain expertise to extract actionable knowledge from vast, complex datasets. Donoho (2017) introduces the concept of “Greater Data Science,” encompassing interconnected activities such as:

- **Data Exploration,**
- **Data Representation,**
- **Computation,**
- **Modeling,**

- **Visualization**, and
- **Meta-science** — the study of data science itself.

This framework underlines Data Science’s role in facilitating ML and AI by preparing and contextualizing data while providing interpretive insights vital for informed decision-making. [2] Kitchin (2014) frames Data Science as emerging from new epistemologies shaped by the explosion of data availability and technological advances in algorithms and infrastructure, making it a centerpiece for scientific, commercial, and societal applications. [9] With the individual roles of AI, ML, and Data Science established, their interconnected nature becomes evident. Data Science provides the data foundation, ML extracts patterns and builds predictive models, and AI integrates these models into intelligent decision-making systems. Yet, despite this synergy, each field maintains its own methodological focus and scope. Recognizing both their overlap and independence is essential for understanding how they function collectively while contributing uniquely to modern data-driven innovation.

The interrelationship among these fields is both collaborative and hierarchical. Data Science primarily handles data acquisition, cleaning, and exploratory analysis, producing structured datasets that ML utilizes for predictive modeling and pattern discovery. AI then integrates these models to build intelligent systems capable of autonomous operation in

complex environments. Marr (n.d.) highlights that tools like scikit-learn and TensorFlow are common across the disciplines, serving as shared frameworks for algorithm development and deployment. [10] However, distinctions remain clear: Data Science excels at insight generation and visualization, ML prioritizes optimization and learning at scale, and AI pursues higher-level reasoning, planning, and interaction. The nested set perspective offered by Magnimind Academy (n.d.) illustrates this conceptual layering, where ML is subsumed under AI, both relying fundamentally on Data Science for data foundation. This nuanced interplay indicates that while the three fields promote synergistic innovation in technology ecosystems, each maintains methodological independence that guides its specific focus and toolsets. [11]

Tools	Category	Primary Use
Python	Data Science/ AI / ML	General-purpose programming, data processing, and model development
NumPy	Data Science	Numerical Computing and Matrix Operations
Pandas	Data Science	Data cleaning, manipulation, and analysis
Scikit-Learn	ML	Classical ML algorithms (classification, regression, clustering)
TensorFlow	ML / AI	Deep learning model building and deployment
PyTorch	ML / AI	Research-friendly deep learning framework
Keras	ML	High-level neural network API built on TensorFlow
Matplotlib / Seaborn	Data Science	Data visualization and exploratory analysis

Table 1. Core Tools and Their Primary Uses in AI, ML, and Data Science.

Analysis

Name of Paper	Authors	Key Findings
A Comparative Analysis of AI, Machine Learning, and Data Science	Prof. Sesham Anand	• Defines AI, ML, and Data Science with clear technical boundaries. • Explains hierarchical structure (Data Science → ML → AI). • Highlights the collaborative nature across data pipelines. • Shows how Data Science prepares data, ML builds models, and AI makes automated decisions. • Emphasizes real-world applications and technological dependencies.
A Survey on AI and ML Techniques	1. Rishikesh Nandankar 2. Rutvik Raut	• Provides an overview of major AI and ML techniques. • Shows how ML algorithms contribute directly to AI advancements. • Describes supervised/unsupervised learning roles in AI systems. • Highlights strengths and limitations of various ML approaches. • Useful for understanding method-level overlap between AI and ML.
Systematic Analysis of Deep Learning Models Compared to Machine Learning	1. Sheshang Degadwala 2. Dhairya Vyas	• Compares ML methods with Deep Learning architectures. • Describes situations where classical ML

		outperforms DL (small, structured data). • Identifies DL strengths in large, unstructured datasets (images, text). • Discusses computational complexities and resource needs. • Positions DL as a subset of ML within broader AI systems.
Trusted AI in Multi-agent Systems: Privacy and Security for Distributed Learning	Chuan Ma et al.	• Examines AI/ML in distributed multi-agent environments. • Highlights privacy and security challenges in collaborative model training. • Explains federated learning and secure aggregation methods. • Demonstrates real-world overlap of AI, ML, and DS in networking systems. • Shows the importance of trust, robustness, and data governance.
The Discipline of Machine Learning	Tom M. Mitchell	• Provides the formal definition of ML used in modern research. • Explains the scope, goals, and methodologies of ML. • Distinguishes ML from traditional programming. • Shows how ML fits within the larger field of AI. • Offers classification of ML algorithms (supervised, unsupervised, RL).
50 Years of Data Science	David Donoho	• Defines the six components of Greater Data Science.

		• Explains how Data Science differs from statistics, ML, and AI. • Shows DS as a broad ecosystem beyond predictive modeling. • Emphasizes DS roles in data prep, exploration, visualization, and interpretation. • Clarifies DS independence despite its integration with ML and AI.
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Conclusion

The examination of Artificial Intelligence, Machine Learning, and Data Science reveals a layered and interdependent technological landscape in which each field contributes distinctly to modern computational intelligence. The literature consistently demonstrates that Data Science forms the analytical foundation by acquiring, cleaning, organizing, and interpreting data—processes that are essential for meaningful model development. Building on this foundation, Machine Learning introduces the ability to learn patterns and make predictions through data-driven algorithms, functioning as the operational core of many intelligent systems. Artificial Intelligence then extends these capabilities further by integrating ML models into systems designed to reason, adapt, and act autonomously in complex environments.

Despite their close relationship, it is equally clear that AI, ML, and Data Science retain independent methodological identities. Data Science emphasizes statistical reasoning and domain-driven insights; Machine Learning focuses on algorithmic learning and optimization; and AI pursues broader goals of cognition, autonomy, and decision-making. This interplay of overlap and independence forms a complementary ecosystem in which each field enhances the capacities of the others while maintaining its specialized contributions. The studies reviewed collectively highlight that their coexistence is not a matter of redundancy but of synergy. The hierarchical nature of these fields—supported by shared tools, evolving research frameworks, and interdisciplinary applications—demonstrates how innovation emerges when statistical analysis, algorithmic learning, and intelligent automation operate together. Recognizing these boundaries and synergies is essential not only for academic clarity but also for designing,

implementing, and evaluating modern data-driven solutions. Ultimately, the relationship among AI, ML, and Data Science reflects a dynamic and evolving technological framework, one that will continue to expand as data grows, algorithms advance, and intelligent systems become increasingly integrated into real-world environments.

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